

29 Aug 2024 01:50:57 ET | 18 pages

NVIDIA Corp (NVDA.O)

Beat & Raise; Stock down on Gross Margin Mix; Maintain Buy

CITI'S TAKE

NVDA reported “beat and raise” results with Jul-Q sales of \$30B towards high-end of \$29-\$30B and Oct-Q guide of \$32.5B at the mid-point of \$32-\$33B expectations. Management sounded confident in Blackwell ramp with production in the Oct-Q and several billion sales in the Jan-Q. However, the impact of higher H200/HBM mix lowers gross margins in the next two quarters. Net-net, our Jan-Q EPS is unchanged (Street likely comes down on lower gross margins) and we maintain above consensus FY26/27 EPS. We expect the stock to likely remain range bound through the next two quarters before Blackwell driven Y/Y sales and gross margin inflection in the Apr-Q. CES Jan is the next major catalyst for the stock. AI adoption remains in 3rd/4th innings as enterprise AI demand takes off with AI agents. NVDA sees 10-20x more data compute long-term. Maintain buy and \$150 TP on consistent ~35x P/E times CY25 EPS.

Blackwell: Management clarified that there are no Blackwell units expected in Oct-Q (only samples). Regarding the mask issues for Blackwell, management mentioned it moved shipments a couple of weeks. They nevertheless expect several \$billions in 4Q. Because they are still supply-constrained, they expect Hopper demand to remain strong for both CSPs and enterprises.

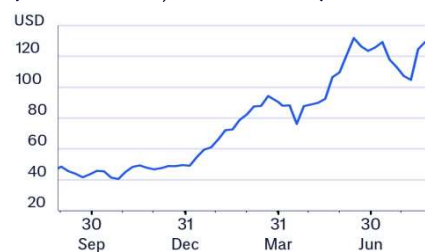
Non-GAAP Gross Margins: Jul-Q and Oct-Q margins coming down is mostly due to H200 having higher memory and therefore cost per GPU thus pulling down NGGM as it ramps. Regarding Jan-Q, management highlighted Blackwell as the key driver to margins likely coming down (we estimate 200 bps+). Nevertheless, Apr-Q margins will depend on how Nvidia will determine cost/price of the scaled-up Blackwell systems which contains GPUs, CPUs, NV links etc.

Buy

Price (28 Aug 24 16:00)	US\$125.61
Target price	US\$150.00
Expected share price return	19.4%
Expected dividend yield	0.0%
Expected total return	19.4%
Market Cap	US\$3,090,006M

Price Performance

(RIC: NVDA.O, BB: NVDA US)


Atif Malik^{AC}

Papa Sylla

EPS (US\$)	Q1	Q2	Q3	Q4	FY	FC Cons	VA Cons
2024A	0.11A	0.29A	0.40A	0.53A	1.31A	1.30A	1.30A
2025E	0.64A	0.72A	0.73E	0.70E	2.68E	2.75E	2.84E
Previous	0.64A	0.65E	0.72E	0.70E	2.61E	na	na
2026E	0.89E	1.00E	1.16E	1.29E	4.31E	3.81E	4.09E
Previous	0.88E	1.00E	1.16E	1.29E	4.30E	na	na
2027E	1.32E	1.37E	1.40E	1.38E	5.51E	4.46E	3.77E
Previous	1.32E	1.37E	1.40E	1.39E	5.51E	na	na

Source: Company Reports and dataCentral, Citi Research. FC Cons: First Call Consensus. VA Cons: Visible Alpha Consensus.

Click [here](#) for Visible Alpha consensus data

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

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NVDA.O: Fiscal year end 31-Jan						Price: US\$125.61; TP: US\$150.00; Market Cap: US\$3,090,006m; Recomm: Buy					
Profit & Loss (US\$m)	2023	2024	2025E	2026E	2027E	Valuation ratios	2023	2024	2025E	2026E	2027E
Sales revenue	26,974	60,922	121,589	198,329	262,284	PE (x)	na	95.9	46.8	29.2	22.8
Cost of sales	-11,618	-16,621	-30,644	-51,986	-69,701	PB (x)	na	72.1	46.7	23.5	13.8
Gross profit	15,356	44,301	90,945	146,344	192,583	EV/EBITDA (x)	na	80.2	37.8	22.4	16.3
Gross Margin (%)	56.9	72.7	74.8	73.8	73.4	FCF yield (%)	0.1	0.9	1.8	2.7	3.8
EBITDA (Adj)	13,339	38,407	80,967	135,065	181,328	Dividend yield (%)	0.1	0.1	0.1	0.1	0.1
EBITDA Margin (Adj) (%)	49.5	63.0	66.6	68.1	69.1	Payout ratio (%)	48	12	6	4	3
Depreciation	-1,490	-1,454	-1,655	-1,768	-1,863	ROE (%)	17.9	92.5	117.2	103.5	74.0
Amortisation	-54	-54	-54	-54	-54	Cashflow (US\$m)	2023	2024	2025E	2026E	2027E
EBIT (Adj)	9,040	37,134	79,902	133,183	179,352	EBITDA	7,122	34,480	76,361	128,188	173,472
EBIT Margin (Adj) (%)	33.5	61.0	65.7	67.2	68.4	Working capital	-2,197	-3,723	-4,055	-14,495	-5,987
Net interest	4	608	1,444	1,532	1,532	Other	726	-2,335	-12,775	-20,148	-28,031
Associates	0	0	0	0	1	Operating cashflow	5,651	28,422	59,531	93,545	139,454
Non-Op/Except/Other Adj	-4,862	-3,924	-4,608	-6,898	-7,876	Capex	-1,833	-1,068	-3,311	-5,950	-7,869
Pre-tax profit	4,182	33,818	76,738	127,818	173,008	Net acq/disposals	9,285	-8,563	-9,346	-3,846	-4,407
Tax	186	-3,727	-11,657	-19,722	-28,585	Other	-78	-971	-627	-467	-576
Extraord./Min.Int./Pref.div.	0	0	0	0	0	Investing cashflow	7,374	-10,602	-13,284	-10,263	-12,851
Reported net profit	4,368	30,091	65,081	108,096	144,424	Dividends paid	0	0	0	0	0
Net Margin (%)	16.2	49.4	53.5	54.5	55.1	Financing cashflow	-11,617	-13,634	-33,152	-30,170	-30,808
Core NPAT	8,366	32,644	67,513	113,002	150,346	Net change in cash	1,408	4,186	13,095	53,112	95,794
Per share data	2023	2024	2025E	2026E	2027E	Free cashflow to s/holders	3,818	27,354	56,220	87,595	131,585
Reported EPS (\$)	0.17	1.21	2.59	4.12	5.29						
Core EPS (\$)	0.33	1.31	2.68	4.31	5.51						
DPS (\$)	0.16	0.16	0.16	0.16	0.16						
CFPS (\$)	0.23	1.14	2.37	3.56	5.11						
FCFPS (\$)	0.15	1.10	2.23	3.34	4.82						
BVPS (\$)	0.90	1.74	2.69	5.35	9.10						
Wtd avg ord shares (m)	24,870	24,693	24,897	25,960	27,014						
Wtd avg diluted shares (m)	25,073	24,933	25,170	26,245	27,310						
Growth rates	2023	2024	2025E	2026E	2027E						
Sales revenue (%)	0.2	125.9	99.6	63.1	32.2						
EBIT (Adj) (%)	-28.8	310.8	115.2	66.7	34.7						
Core NPAT (%)	-25.7	290.2	106.8	67.4	33.0						
Core EPS (%)	-24.9	292.4	104.9	60.5	27.9						
Balance Sheet (US\$m)	2023	2024	2025E	2026E	2027E						
Cash & cash equiv.	13,296	25,984	46,604	99,716	195,509						
Accounts receivables	3,827	9,999	15,523	27,768	32,562						
Inventory	5,159	5,282	8,311	13,898	16,582						
Net fixed & other tangibles	12,061	15,841	14,294	19,620	26,457						
Goodwill & intangibles	6,048	5,542	6,212	6,052	5,934						
Financial & other assets	791	3,080	3,571	3,590	3,579						
Total assets	41,182	65,728	94,515	170,644	280,623						
Accounts payable	1,193	2,699	4,582	7,662	9,142						
Short-term debt	1,250	1,250	0	0	0						
Long-term debt	10,605	9,578	9,765	9,765	9,765						
Provisions & other liab	6,033	9,223	12,067	12,343	12,344						
Total liabilities	19,081	22,750	26,413	29,770	31,250						
Shareholders' equity	22,101	42,978	68,101	140,874	249,373						
Minority interests	0	0	0	0	0						
Total equity	22,101	42,978	68,101	140,874	249,373						
Net debt (Adj)	-1,441	-15,156	-36,839	-89,951	-185,744						
Net debt to equity (Adj) (%)	-6.5	-35.3	-54.1	-63.9	-74.5						
For definitions of the items in this table, please click here .											

NVDA Jul-Q Review

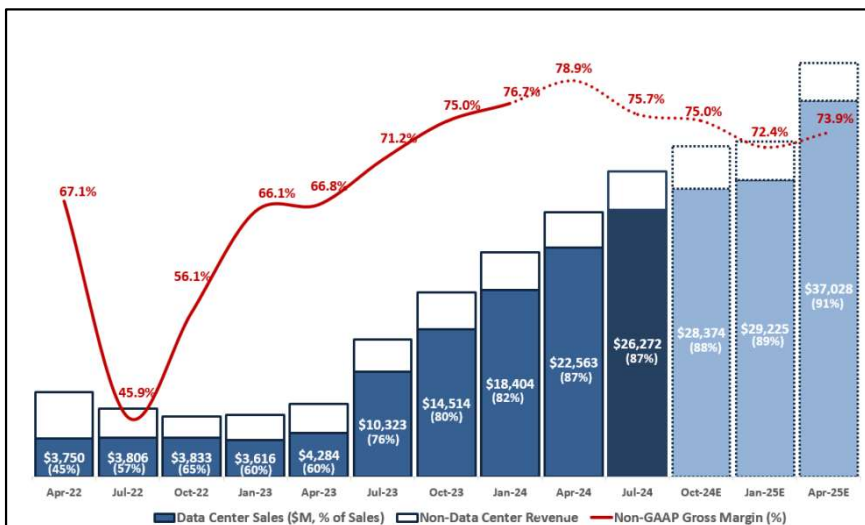
I- Results and Outlook

NVDA reported Jul-Q revenue of \$30B, +15% Q/Q and +122% Y/Y. Non-GAAP EPS (ex-SBC) was \$0.68, +11% QoQ/+153% YoY (post stock-split). Jul-Q non-GAAP Gross Margin ex-SBC (NGGM) was down (as expected) 300+bps to 75.7%.

Going by segments:

- **Data Center (DC) sales** rose 16% Q/Q (vs. Citi's/Street's +10%/+11%) boosted by the continued strong demand of AI products, led notably by the Hopper platform with H100 demand remaining strong and the H200 successfully launching this quarter. The segment represented 87% of the quarter's total sales, flat vs. Jul-Q. Management expects Hopper to continue to grow in 2H amid strong demand and improving supply. After a sequential drop in Apr-Q amid supply timing issues (likely optics), **the Networking business** was up sequentially (+16% and up 2x YoY). Within Networking, AI Ethernet including Spectrum-X, doubled QoQ with hundreds of customers adopting NVDA's Ethernet offerings. Management adds that Spectrum-X has broad market support from OEM and ODM partners and is being adopted by CSPs, GPU cloud providers and enterprises. The company added that **in terms of DC end-markets**, cloud service providers represented 40%+ of sales and enterprises, consumer internet companies and other non-hyperscalers markets representing a combined 50%+ of NVDA's data center revenue in July. We think a strong enterprise, sovereign and consumer internet sales mix bodes well with NVDA's push to expand its offering beyond the hyperscalers. **Inference** workload sales which have surprised positively back in January remains similar to the Jul-Q at the ~40% levels as NVDA's inference platform throughput and efficiency continue to strongly attract enterprises in particular. The company highlighted its latest and leading MLPerf inference results for both Hopper and Blackwell.

Figure 1. Data Center Sales Growth/Mix Trends



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*Note: Non-GAAP EPS excludes SBC-Citi data central EPS includes SBC.

Source: Citi Research, Company Reports

Double clicking on Blackwell... NVDA emphasized that the platform will see “several” \$billions in sales in Q4 (Jan-Q). The company mentioned demand running well above supply and expects that trend to persist. Regarding the change in design to Blackwell, Jensen discussed the changes to be around the GPU mask without any need for functional changes. Blackwell, Grace-Blackwell and all the remaining 100+ Blackwell-based systems are sampling currently. He reiterated that Blackwell will start production in Q4.

Software, SaaS and support sales is approaching \$2B annual run-rate exiting this year with NVIDIA AI Enterprise being a contributor to the growth. In fact, the company’s latest AI software platform NIMs have been adopted by over 150 partners the company highlights. The adoption has been broad going from healthcare to retail with some companies saving up to 70% of cost and 8x latency reduction using NIMs. This further consolidates our view that AI Enterprise which is the next big frontier for NVDA in terms of end-market demand with AI agents such as NVDA’s latest (NIM Agent Blueprint) as the primary enabler.

Pivoting to China DC sales, NVDA flagged a sequential uptick in Jul-Q following a significant China sales QoQ declines in both Jan-Q and Apr-Q following the U.S. government export control regulations imposed in October. The company discussed the market being more competitive going forward.

- **Gaming revenue** was up 9% Q/Q (vs. Citi’s/Street’s +4%/+4%) and up 16% YoY. The segment represented 10% of sales flat vs. Apr-Q. Gaming growth was increased demand in console, notebook, and desktop revenue. We continue to model the segment being in lockstep with notebook seasonality, so being up MSD in Oct-Q. The company also mentioned that every PC with RTX is an AI PC. RTX PCs can deliver up to 1,300 AI tops and are now over 200 RTX AI laptops designed from leading PC manufacturers. Recall at CES, that their new 40 series PC GPUs RTX 40 SUPER and applications such as NVIDIA AI Workbench as an app can be important in that trend. The app is a unified, easy-to-use toolkit that allows developers to quickly create, test, and customize pre-trained generative AI models and LLMs. NVDA estimates that their 100M+ RTX PC installed base offer them the perfect platform to play a significant role in the emerging AI PC market.

- **Pro Visualization revenue** was up 6% QoQ (vs. Citi’s/Street’s +8%/+5%) and 20% YoY. The sequential and YoY growths were driven by AI and graphic use cases with auto and manufacturing verticals among the key drivers. The sales uptick reflects a richer mix of RTX Ada architecture GPUs. Additionally, Enterprises are refreshing their workstations to support generative AI-related workloads such as data preparation, LLM fine-tuning and retrieval augmented generation which bodes well for the segment. Longer term, NVDA estimates that Gen AI and Omniverse industrial digitalization will drive the next wave of growth for the segment.

- **Auto sales** rose a more modest 5% Q/Q (vs. Citi’s/Street’s +10%/+5%) when compared to Apr-Q’s 17%. That said YoY was an impressive 37%. The growth was driven by a new customer ramp in self-driving platforms as well as a continued growth of NVDA’s cockpit solutions and self-driving platforms. To us it is clear NVDA is seeing growth in in-car chip demand as some customers are starting to launch L3 which requires more advanced algorithms as well as larger models. Outside the vehicle they are seeing a large opportunity with, for instance, their digital twin offering with Omniverse used by OEMs to test their AV applications. At CES, NVDA discussed how AI is everywhere now, hence the need for additional

products to meet that demand. The new 1-year cadence will include new products with new architectures as well as ones based on pre-existing architecture. They believe the variety of products will help NVDA meet the demand of various entities. Recently, NVDA announced multiple design wins for its upcoming NVIDIA Drive Thor products. The chips are based on the new NVIDIA Blackwell architecture and has several leading EV makers planning to start their EV product using Thor next year next year.

Non-GAAP GM (ex-SBC) as expected came down in Jul-Q but remained 20 bps above the company’s 75.5% guidance (at MP). Citi and Visible Alpha’s estimates were 75% and 75.6% respectively. The non-GAAP GM (ex-SBC) downtick reflects the increased mix of new data center products (with higher HBM memory and so cost) as data center sales mix was flat QoQ at 87%. When it comes to FY2025, the company continues to guide to mid-70s% as the Blackwell family of chips will go through higher cost before moderating when shipment at scale, cost optimizations and larger systems sales take effect. The Oct-Q’s 75% (at MP) and the FY2025 mid 70s% guides, imply a 200+bps NGGM (ex-SBC) decline in Jan-Q. **Non-GAAP operating expenses (ex-SBC)** were \$2.8B (ex-SBC), up 12% QoQ, reflecting primarily higher compensation-related cost. Non-GAAP net income (ex-SBC) positively surprised by 8% vs. Citi’s \$15B estimates to reach \$16.9B. **Non-GAAP EPS (ex-SBC)** sat at \$0.68, up 11% QoQ and 153% YoY (taking into account share-split pro forma). Citi’s and Street’s non-GAAP EPS (ex-SBC) were \$0.63 and \$0.64 respectively.

Figure 2. Summary of Jul-Q Results vs. Guidance and Expectations*

F2Q25- Jul	Guidance	Citi Est	Results	Street (Prior to Results)
Revenue (\$M)	\$28,000 +/- 2%	\$28,484	\$30,040	\$28,653
Sequential growth	8%	9%	15%	10%
EPS (non-GAAP)		\$0.63	\$0.68	\$0.64
Gross Margin (non-GAAP)	75.5% +/- 50bps	75.0%	75.7%	75.6%
OpEx (non-GAAP, \$M)	\$2,800	\$2,876	\$2,792	
Operating Margin (non-GAAP)		64.5%	66.4%	65.9%
Tax rate	17% +/- 1%	17%	17%	

Consensus as of: 8/28/24
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*Note: Non-GAAP EPS excludes SBC-Citi data central EPS includes SBC.
Source: Citi Research, FactSet

Oct-Q/FY2025 guide: NVDA guided Oct-Q revenue to \$32.5B at MP, up 8% Q/Q and 79% Y/Y. Prior to results, Street expected \$31.5B (FactSet). Data center growth, yet again, is expected to be the primary driver of the sequential sales growth backed notably by the growing Gen AI demand of AI GPUs and networking products particularly NVDA’s Ethernet platform. The company guided to a 75% NGGM in Oct-Q 140 bps above our expectations but 30bps Street’s (FactSet). For FY2025, NVDA expect NGGM to land in the mid-70% range. Non-GAAP operating expenses are guided to sum-up to \$3B in Jul-Q up 7% Q/Q amid sales expansion. FY2025 Non-GAAP operating expenses are expected to grow in the mid-to-upper 40% (revised up from low 40%) as the company works on developing its next-gen products.

Figure 3. Summary of Oct-Q Guidance vs. Expectations *

F3Q25E- Oct	Guidance	Citi Old	Citi New	Street (Prior to Results)
Revenue (\$M)	\$32,500 +/- 2%	\$31,953	\$32,508	\$31,540
Sequential growth	8%	12%	8%	10%
EPS (non-GAAP)		\$0.70	\$0.71	\$0.71
Gross Margin (non-GAAP)	75% +/- 50bps	73.6%	75.0%	75.4%
OpEx (non-GAAP, \$M)	\$3,000	\$3,006	\$3,068	
Operating Margin (non-GAAP)		64.2%	65.6%	65.9%
Tax rate	17% +/- 1%	17%	17%	

Consensus as of: 8/28/24

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*Note: Non-GAAP EPS excludes SBC-Citi data central EPS includes SBC.

Source: Citi Research, FactSet

Figure 4. Annual P&L Changes (FY2020-FY2027E) *

Key Financial Data -- Income Statement	FY20	FY21	FY22	FY23	FY24	Old FY25E	New FY25E	Old FY26E	New FY26E	Old FY27E	New FY27E
(\$M) except per share	Jan-20	Jan-21	Jan-22	Jan-23	Jan-24	Jan-25	Jan-25	Jan-26	Jan-26	Jan-27	Jan-27
Revenue (\$M)	10,318	16,675	26,914	26,974	60,922	118,087	121,589	197,066	198,329	260,700	262,284
YY	-6.8%	52.7%	61.4%	0.2%	125.9%	93.8%	99.6%	66.9%	63.1%	32.3%	32.2%
	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Gross Margin (ex-1, inc-SBC)	62.1%	65.1%	66.2%	58.7%	73.6%	74.8%	75.2%	73.2%	74.0%	73.0%	73.6%
RD % sales	30.6%	28.7%	24.4%	34.2%	18.4%	13.3%	13.3%	9.6%	10.1%	7.7%	8.1%
SG&A % sales	12.5%	14.3%	10.2%	11.6%	5.8%	3.8%	3.8%	3.0%	3.0%	2.5%	2.6%
OpEx (ex-1, inc-SBC) % Sales	28.3%	24.9%	19.6%	25.7%	12.8%	9.8%	9.6%	6.8%	7.0%	5.2%	5.3%
R&D	2,828.0	3,926.0	5,267.0	7,338.0	8,674.0	12,530.0	12,790.0	14,880.0	15,360.0	15,280.0	15,960.0
SG&A	1,094.0	1,938.0	2,166.0	2,440.0	2,655.0	3,453.0	3,503.0	4,418.0	4,618.0	4,868.0	5,068.0
OpEx (ex-1, inc-SBC)	3,922.0	5,864.0	7,433.0	9,778.0	11,329.0	15,983.0	16,293.0	19,098.0	19,978.0	20,148.0	21,028.0
Operating Income (ex-1, inc-SBC)	2,861.0	4,995.0	10,395.0	6,049.0	33,488.0	72,321.2	75,118.1	125,101.6	126,829.6	170,056.5	172,019.5
Operating Margin (ex-1, inc-SBC)	26.2%	30.0%	38.6%	22.4%	55.0%	61.2%	61.8%	63.5%	63.9%	65.2%	65.6%
Tax (ex-1, inc-SBC)	280.0	412.0	1234.0	680.0	5101.0	13046.9	13831.1	21526.6	21692.7	30318.5	30518.5
Tax Rate (ex-1, inc-SBC)	9.4%	8.5%	12.0%	11.3%	14.9%	17.7%	17.9%	17.1%	16.9%	17.7%	17.6%
Share Count	24710.0	25090.0	25357.5	25072.5	24932.5	24888.7	25170.3	25771.1	26244.8	26817.5	27310.4
EPS (ex-1, inc-SBC)	\$0.11	\$0.18	\$0.36	\$0.21	\$1.17	\$2.44	\$2.52	\$4.06	\$4.06	\$5.25	\$5.23
EPS (ex-1, ex-SBC)	\$0.14	\$0.25	\$0.44	\$0.33	\$1.31	\$2.61	\$2.68	\$4.30	\$4.31	\$5.51	\$5.51
Street Revs							121,277		168,334		196,887
Street EPS (ex-1, ex-SBC)							\$2.73		\$3.81		\$4.46

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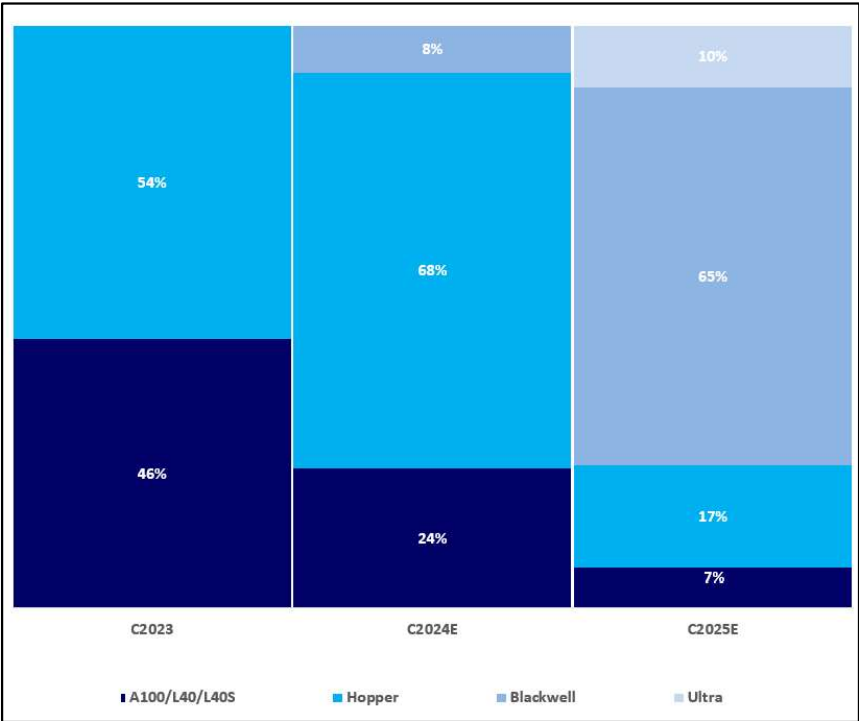
Source: Citi Research, FactSet

II-NVDA AI GPU Model

Units... We model 5M (+92% Y/Y) and 5.9M (+28% Y/Y) units in FY2025/FY2026. Regarding Hopper, we previously raised more H100/H200 as we expect some customers to turn to H200 (particularly in Oct-Q) amid Blackwell delays. We are modeling 3.4M/1.2M Hopper GPUs in FY2025 and FY2026. In FY2026, we are estimating ~75K NVL 36 equivalent servers (both NVL36*2 and NVL72 converted to NVL 36 format), which represents 2.7M GPUs and 73% of our total Blackwell deployments estimates in FY2026.

AI GPU sales: Backed by units' upticks and favorable product mix, we estimate FY2025/FY2026 sales reaching \$88B (+83%)/\$162B (+84%), respectively. We expect sales of AI GPUs to represent on average mid-to-high 80% of total NVDA data center sales in FY2025-FY2026, with the remaining being comprised primarily of networking (Ethernet and InfiniBand), standalone CPUs, and software/services offerings.

Figure 5. NVDA AI GPU Units Mix



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Source: Citi Research

Figure 6. NVDA AI GPU Scenario Model

Bear Scenario									
	Fiscal 2024			Fiscal 2025			Fiscal 2026		
	Quarterly	Annual	ASP (\$K)	Quarterly	Annual	ASP (\$K)	Quarterly	Annual	ASP (\$K)
H100/H200 Based Units (K)	350	1,400	\$28	550	2,200	\$17	250	1,000	\$10
A100/L40/L40S* Based Units (K)	300	1,200	\$8	300	1,200	\$5	150	600	\$5
B100/B200 Based Units (K)	-	\$0		120	480	\$30	800	3,200	\$25
Ultra Based Units (K)		\$0		-	-	\$0	250	1,000	\$30
Total GPUs Units (K)	650	2,600	\$19	970	3,880	\$15	1,450	5,800	\$21
YoY					49%			49%	
H100/H200 Based Sales (\$M)		\$39,200			\$37,400			\$10,000	
A100/L40/L40S* Based Sales (\$M)		\$9,000			\$6,000			\$3,000	
B100/B200 Based sales (\$M)		\$0			\$14,400			\$80,000	
Ultra Based sales (\$M)		\$0			\$0			\$30,000	
Total GPUs Sales (\$M)		\$48,200			\$57,800			\$123,000	
YoY		281%			20%			113%	

Base Scenario									
	Fiscal 2024			Fiscal 2025			Fiscal 2026		
	Quarterly	Annual	ASP (\$K)	Quarterly	Annual	ASP (\$K)	Quarterly	Annual	ASP (\$K)
H100/H200 Based Units (K)	350	1,400	\$28	850	3,400	\$20	300	1,200	\$10
A100/L40/L40S* Based Units (K)	300	1,200	\$8	300	1,200	\$5	100	400	\$4
B100/B200 (B200A) Based Units (K)	-	-	\$0	100	400	\$32	930	3,720	\$3
Ultra Based Units (K)	-	-	\$0	-	-	\$0	150	600	\$3
Total GPUs Units (K)	650	2,600	\$19	1,250	5,000	\$18	1,480	5,920	\$2
YoY					92%			18%	
H100/H200 Based Sales (\$M)		\$39,200			\$69,360			\$12,000	
A100/L40/L40S* Based Sales (\$M)		\$9,000			\$6,000			\$1,400	
B100/B200 Based sales (\$M)		\$0			\$12,800			\$128,340	
Ultra Based sales (\$M)		\$0			\$0			\$20,400	
Total GPUs Sales (\$M)		\$48,200			\$88,160			\$162,140	
YoY		281%			83%			84%	

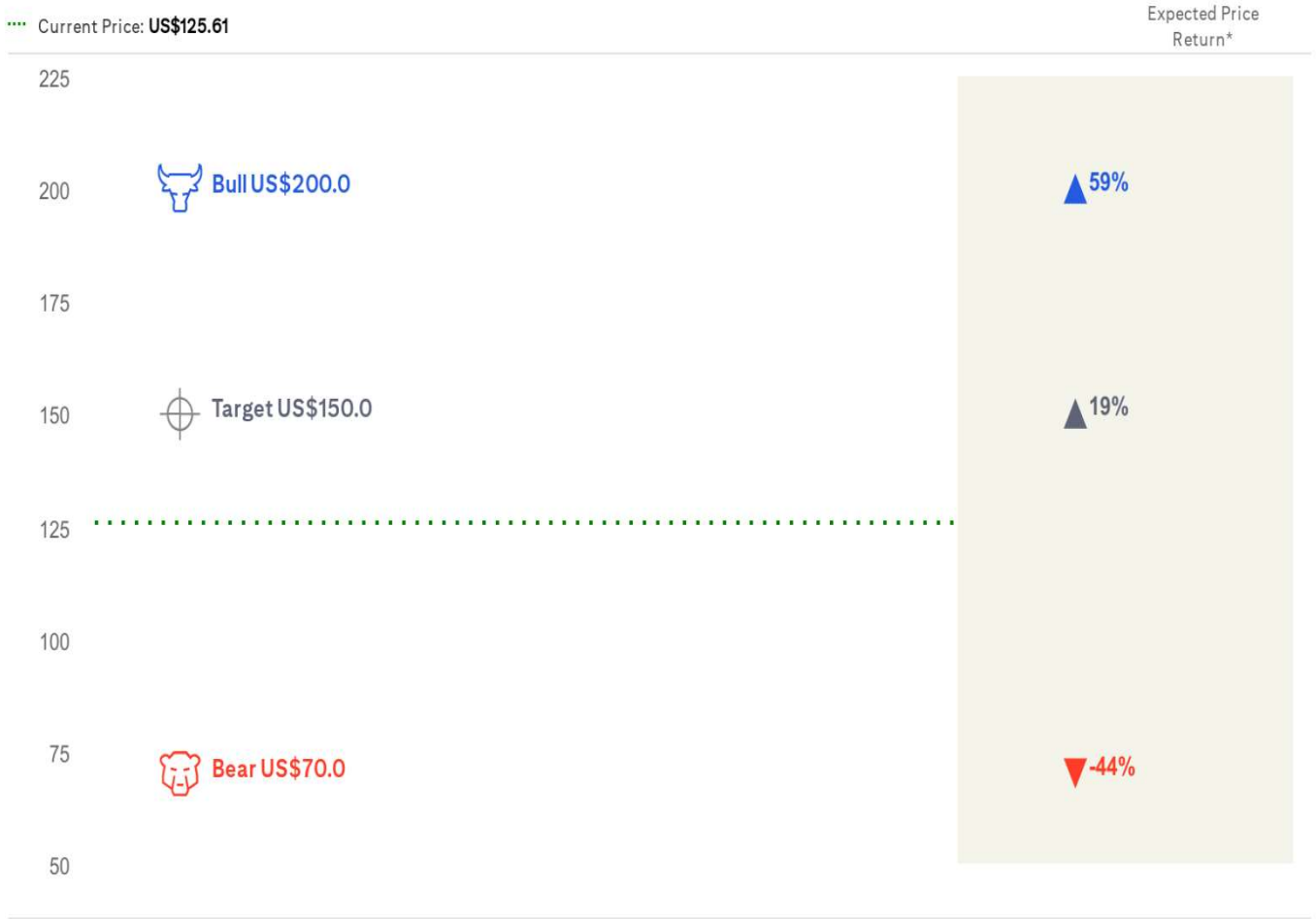
Bull Scenario									
	Fiscal 2024			Fiscal 2025			Fiscal 2026		
	Quarterly	Annual	ASP (\$K)	Quarterly	Annual	ASP (\$K)	Quarterly	Annual	ASP (\$K)
H100/H200 Based Units (K)	350	1,400	\$28	800	3,200	\$25	320	1,280	\$10
A100/L40/L40S* Based Units (K)	300	1,200	\$8	300	1,200	\$8	250	1,000	\$5
B100/B200 Based Units (K)	-	-	\$0	150	600	\$37	1,000	4,000	\$34
Ultra Based Units (K)	-	-	\$0	-	-	\$0	350	1,400	\$37
Total GPUs Units (K)	650	2,600	\$19	1,250	5,000	\$22	1,920	7,680	\$27
YoY					92%			54%	
H100/H200 Based Sales (\$M)		\$39,200			\$80,000			\$12,800	
A100/L40/L40S* Based Sales (\$M)		\$9,000			\$9,000			\$5,000	
B100/B200 Based sales (\$M)		\$0			\$22,200			\$136,000	
Ultra Based sales (\$M)		\$0			\$0			\$51,800	
Total GPUs Sales (\$M)		\$48,200			\$111,200			\$205,600	
YoY		281%			131%			85%	

* Includes potential product upgrades.

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Source: Citi Research

Bull/Bear: NVIDIA Corp (NVDA.O)



Spread* 103pp
*Current Price and expected returns as of 28 August 2024

BULL Assumptions

40x P/E
AI compute TAM of \$300B by 2027

BASE Assumptions

35x P/E
AI compute TAM of \$200B by 2027

BEAR Assumptions

25x P/E
AI compute TAM of \$100B by 2027 on higher competition/pricing pressure.

NVIDIA Corp

Company description

Nvidia is a leader in graphics processing units or GPUs. It is engaged in creating graphics and networking chips, which are used in gaming, data centers, and automotive end markets. Nvidia was founded by Jen-Hsun Huang (current CEO) and partners in January 1993 and is headquartered in Santa Clara, CA.

Investment strategy

We like Buy-rated NVDA on secular AI growth opportunities.

Valuation

Our price target for NVDA is \$150 based on ~35x P/E on C25E EPS. Our 35x P/E multiple is roughly in-line with the one-year average but below 3-year and 5 year averages.

Risks

Downside risks to the attainment of our target price include: 1) competition on gaming could drive the stock lower if Nvidia loses market share; 2) slower-than-expected adoption of new platforms can drive lower data center and gaming sales; 3) lumpiness in auto and data center markets can add volatility to the stock/multiple; and 4) cryptomining impact on gaming sales.

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Appendix A-1

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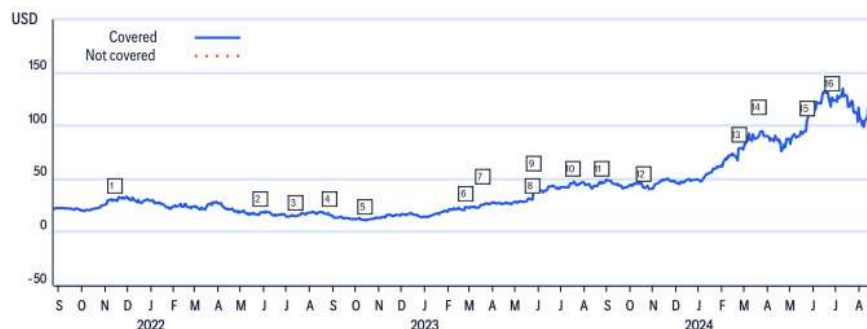
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NVIDIA Corp (NVDA)

Ratings and Target Price History
Fundamental Research

Analyst: Atif Malik



	Date	Rating	Target Price	Closing Price
1	14-Nov-21 21:33:02	1	*35.00	30.39
2	26-May-22 02:00:53	1	*31.50	17.85
3	11-Jul-22 19:37:43	1	*28.50	15.15
4	25-Aug-22 02:27:54	1	*24.80	17.91
5	11-Oct-22 05:17:28	1	*21.00	11.59
6	23-Feb-23 02:35:10	1	*24.50	23.66

*Indicates Change

	Date	Rating	Target Price	Closing Price
7	17-Mar-23 06:01:34	1	*30.50	25.73
8	23-May-23 08:08:33	1	*36.30	30.69
9	25-May-23 02:22:12	1	*42.00	37.98
10	17-Jul-23 01:50:06	1	*52.00	46.46
11	24-Aug-23 02:36:33	1	*63.00	47.16
12	18-Oct-23 00:32:40	1	*57.50	42.20

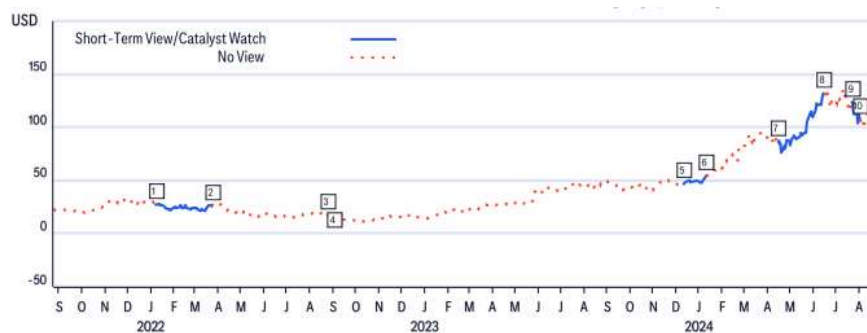
	Date	Rating	Target Price	Closing Price
13	22-Feb-24 02:01:24	1	*82.00	78.54
14	20-Mar-24 06:26:15	1	*103.00	90.37
15	23-May-24 02:55:03	1	*126.00	103.80
16	26-Jun-24 05:00:00	1	*150.00	126.40

Rating/target price changes above reflect Eastern Time

NVIDIA Corp (NVDA)

Short-Term View/Catalyst Watch Research

Analyst: Atif Malik



	Date	Action	Expected Direction	Duration	Closing Price
1	07-Jan-22 00:00:00	Add CW	Upside	90 Days	27.25
2	23-Mar-22 01:00:00	Remove CW	Upside	90 Days	25.63
3	24-Aug-22 22:27:54	Add CW	Upside	30 Days	17.22
4	01-Sep-22 02:57:28	Remove CW	Upside	30 Days	13.94

CW - Catalyst Watch, STV - Short-Term View

	Date	Action	Expected Direction	Duration	Closing Price
5	11-Dec-23 08:11:55	Add CW	Upside	30 Days	46.63
6	10-Jan-24 11:06:14	Remove CW	Upside	30 Days	54.35
7	15-Apr-24 01:00:00	Add CW	Upside	90 Days	86.00
8	16-Jun-24 21:00:00	Remove CW	Upside	90 Days	131.88

	Date	Action	Expected Direction	Duration	Closing Price
9	22-Jul-24 01:00:00	Add CW	Upside	30 Days	123.54
10	04-Aug-24 14:08:32	Remove CW	Upside	30 Days	107.27

Rating/target price changes above reflect Eastern Time

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Data current as of 01 Jul 2024	12 Month Rating			Catalyst Watch		
	Buy	Hold	Sell	Buy	Hold	Sell
Citi Research Global Fundamental Coverage (Neutral=Hold)	58%	33%	9%	34%	50%	16%
% of companies in each rating category that are investment banking clients	48%	50%	36%	51%	45%	46%

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